RAJALAKSHMI ENGINEERING COLLEGE

V Jikshitha 251801118-2025-AIDS V2

Started on	Saturday, 12 September 2026, 2:28 PM
State	Finished
Completed on	Saturday, 12 September 2026, 2:30 PM
Time taken	1 min 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int upper_bound(int *arr, int len, int target) {
    int low = 0, high = len;
    while (low < high) {
        int mid = low + (high - low) / 2;
        if (arr[mid] <= target) {
            low = mid + 1;
        } else {
            high = mid;
        }
    }
    return low;
}

int lengthOfLNDS(int *nums, int numsSize) {
    if (numsSize == 0) return 0;

    int *tails = (int *)malloc(numsSize * sizeof(int));
    int len = 0;

    for (int i = 0; i < numsSize; i++) {
        int x = nums[i];

        int idx = upper_bound(tails, len, x);

        tails[idx] = x;

        if (idx == len) {
            len++;
        }
    }

    free(tails);
    return len;
}

int main() {
    int n;


    if (scanf("%d", &n) != 1) return 0;

    int *sequence = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &sequence[i]);
    }

    printf("%d\n", lengthOfLNDS(sequence, n));

    free(sequence);
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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